

“Combining graph databases and LLMs can improve the accuracy of LLM results by 54.2%!”

Stephen Chin, VP of Developer Relations at Neo4j



Neo4j's **Stephen Chin** believes that a graph is the most intuitive and efficient data structure for storing, searching, and querying data with complex relationships. In a conversation with **OSFY's Yashasvini Razdan**, the VP of Developer Relations at Neo4j spoke about the applications of graphical databases in AI and LLM-based solutions...

Q What industry trends are driving the need for graph technology?

A. There is a major gap between the number of AI applications successfully put into production and those meeting the intended requirements. The primary challenge lies in whether large language models (LLMs) and applications deliver accurate results, avoid hallucinations, and effectively utilise enterprise data and systems. Knowledge graphs bridge this gap by providing additional context and information, avoiding the 'black box' problem, and providing a smooth development experience without struggling with models that aren't well-suited to the task. Research studies have shown that combining graph databases and LLMs can improve the accuracy of LLM results by 50%. This is significant because one of the biggest hurdles to widespread adoption of LLM technology is ensuring accurate, explainable results that can be confidently used in business contexts, especially for mission-critical applications.

By using knowledge graphs, organisations can transition from AI theories and prototypes to building production-ready applications. These applications not only deliver reliable results but also ensure data meets the needs of both internal and external stakeholders with confidence.

Q So where does open source come into play in all of this?

A. Neo4j has evolved into a highly successful commercial and enterprise database but remains firmly rooted in its open source origins. It is entirely built on open source with an open-sourced database. We have an open source Community Edition, which is free for anyone to use. Without the power of an open source ecosystem, graph technology would never have taken off.

Q What is the advantage of using open source AI solutions compared to proprietary solutions?

A. I would say the ability to construct a best-of-breed AI system. You can deploy it entirely on-premises, host it in the cloud, or use base models like ChatGPT or similar systems. You can opt for open source models that you host yourself, entirely behind your firewall.

This flexibility allows you to select the best components for building your AI solution, tailoring it to meet your specific data privacy requirements, data storage locations, and business needs. An open source graph database enables the creation of an end-to-end application framework to achieve this.

One example of our collaboration in this space is with the Linux Foundation's AI and Data Foundation. We are part of the OPEA project (Open Platform for Enterprise AI), an entirely open source framework for building generative AI applications powered by knowledge graphs.

Q What are the open source software platforms and partnerships that are helping you expand through the community?